# ENGR 1451/2451: Exploratory Data Science

## Midterm Review

Prof. Paul W. Leu
University of Pittsburgh

Fall 2026

# pandas: Inspect, Select, and Summarize

## Typical analysis

```
import pandas as pd
df = pd.read_csv("data.csv")
print(df.shape)
print(df.dtypes)
clean = df.dropna(subset=["group", "value"])
subset = clean.loc[clean["value"] > 0,
                   ["group", "value"]]
summary = (subset.groupby("group")
           ["value"].agg(["count", "mean", "std"]))
```

## What each step does

- Import a CSV
- Inspect dimensions/types
- Remove selected missing entries
- Filter rows; select columns
- Summarize *within* groups

# Population, Sample, and Variables

## Population versus sample

- **Population**: cases about which we want conclusions.
- **Sample**: cases actually observed.
- **Parameter**: population characteristic (e.g., $\mu$, $\sigma$).
- **Statistic**: sample summary (e.g., $\bar{x}$, $s$).

## Variable types

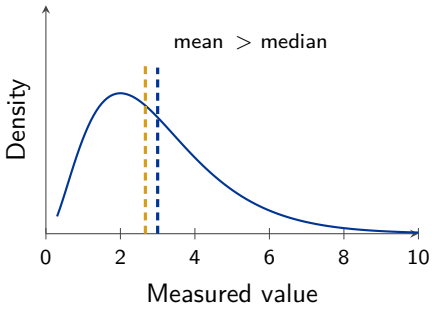
- **Quantitative**: measured numeric value; units matter.
- **Categorical**: groups/labels.
- **Observation**: one row/case (often), not one column.

# Measures of Center and Shape

## Mean versus median

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- Mean uses all values; sensitive to extremes.
- Median is the middle ordered value; resistant to outliers.
- Right-skewed distribution: mean $>$ median.



# Variation: Sample SD and IQR

## Standard deviation

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}, \qquad s = \sqrt{s^2}$$

- Units of $s$: same as the measurements.
- $s^2$ is in squared units.
- Spread around the *mean*; sensitive to outliers.

## Quartiles and outliers

$$\mathrm{IQR} = Q_3 - Q_1$$

Common 1.5 IQR screening fences:

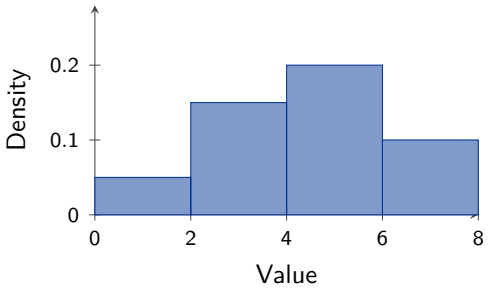
$$Q_1 - 1.5\,\mathrm{IQR}, \quad Q_3 + 1.5\,\mathrm{IQR}.$$

# Histogram Density Is Area, Not Height

## Equal-width bins

$$\text{density height} = \frac{\text{count in bin}}{n \times \text{bin width}}$$

- Bin *area* gives the sample proportion.
- Total area of a density histogram is 1.
- Bin width changes apparent shape.




For unequal-width bins, height still must reflect *density* so bar area is comparable.

# Histogram Area Example

**Find a percentage from a histogram**

On the interval 125–130 mm, the height of the histogram is about 2.1% per mm. What percentage of the women had blood pressures in this class interval?

# Mean Versus Median Example

**An incorrect entry in an income file**

An investigator has a computer file showing family incomes for 1,000 subjects in a certain study. These range from \$5,800 a year to \$98,600 a year. By accident, the highest income in the file gets changed to \$986,000.

(a) Does this affect the average? If so, by how much?

(b) Does this affect the median? If so, by how much?

# What Happens When We Transform Data?

## If each value becomes $y = a + bx$

| Summary | Effect of adding $a$ | Effect of multiplying by $b$ |
|---|---|---|
| Mean, median | Add $a$ | Multiply by $b$ (median too) |
| SD, IQR | Unchanged | Multiply by $\lvert b \rvert$ |
| Variance | Unchanged | Multiply by $b^2$ |
| Standardized $z$ | Unchanged | Same for $b > 0$; sign reverses for $b < 0$ |

## Quick check

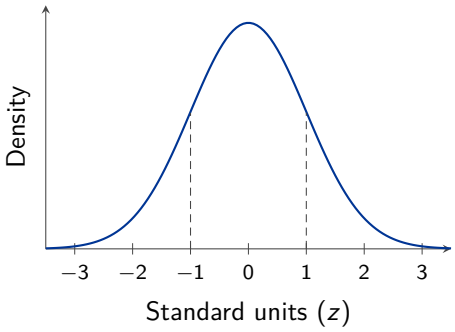
If lengths in centimeters are converted to millimeters, what happens to mean, SD, and variance?

# The Normal Model and Standard Units

## Standardize a measurement

$$z = \frac{x - \mu}{\sigma} \quad \text{or} \quad z = \frac{x - \bar{x}}{s}$$

- $z = +1$: one SD above the reference mean.
- $z = -2$: two SDs below it.
- Normal model uses center and spread to estimate areas.



# Standard Units Example

**Test scores**

On a certain exam, the average of the scores was 50 and the SD was 10.

(a) Convert each of the following scores to standard units: 60, 45, 75.

(b) Find the scores which in standard units are: 0, +1.5, -2.8.

# Normal-Curve Areas and Cutoffs

## Rule of thumb

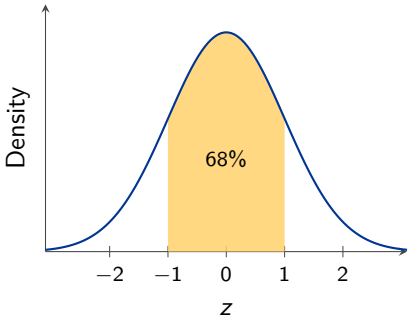
For a normal distribution, approximately:

- 68% within $\pm 1\sigma$
- 95% within $\pm 2\sigma$
- 99.7% within $\pm 3\sigma$

## Exact questions

$$P(a \leq X \leq b) = F(b) - F(a)$$

$F(x)$ is the cumulative distribution function (CDF).



# Empirical Distribution versus Fitted Normal

## Observed data

- Count observations in each bin.
- Compute empirical proportions and quantiles.
- Inspect skew, gaps, multimodality, and outliers.

## Normal approximation

- Estimate center and SD from the sample.
- Approximate probabilities using curve *areas*.
- Evaluate whether the model describes the data adequately.

# Measurements: Chance Error and Bias

## Conceptual model

$$\text{reading} = \text{truth} + \text{bias} + \text{chance error}$$

- **Chance error**: reading-to-reading variability.
- **Bias**: systematic shift of readings.
- Repeatability tells us about scatter
- Calibration or a reference is needed to assess bias.

## Precision is not accuracy

- Precise: readings cluster tightly.
- Accurate: results close to the true value.
- A tight cluster can be consistently wrong.

# Age in the Census Example

## Last digit of a person's reported age

| Last digit of age | 1880 | 1970 |
|---|---|---|
| 0 | 16.8% | 10.6% |
| 5 | 13.4% | 10.0% |

(b) In 1880, there was a strong preference for the digits 0 and 5. How can this be explained?

(c) In 1970, the preference was much weaker. How can this be explained?

# Outliers: Investigate, Do Not Erase

## Potential reasons

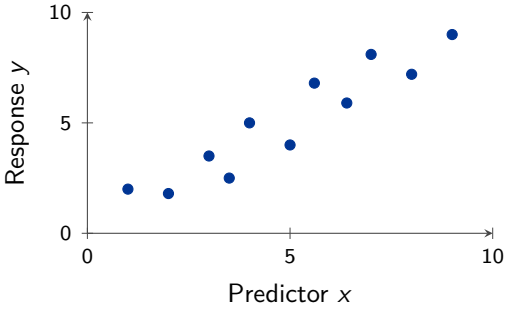
- Recording or unit errors
- Instrument or process failure
- Rare but legitimate outcomes
- A different subgroup or mechanism

## Good reporting

- Plot raw observations.
- Check cause against independent evidence.
- Describe any exclusion criterion.
- Assess sensitivity with and without unusual points.

Removing points solely to make a model look normal or precise is **not** justified.

# Scatterplots: Direction, Form, Strength




**Describe before calculating**

- **Direction**: increasing or decreasing?
- **Form**: straight, curved, clustered?
- **Strength**: tight or diffuse around the trend?
- **Outliers**: any unusual case?

# Correlation: Meaning and Limitations

## Pearson correlation

$$r = \frac{1}{n-1} \sum_{i=1}^{n} \left( \frac{x_i - \bar{x}}{s_x} \right) \left( \frac{y_i - \bar{y}}{s_y} \right)$$

- $-1 \leq r \leq 1$.
- Sign: direction of **linear** association.
- $|r|$: tightness around a straight line.

## What $r$ does not tell us

- Size of a slope or effect in units
- Whether a pattern is nonlinear
- Whether association is causal
- Whether an outlier dominates it

# Correlation Example

**A perfectly proportional relationship**

Suppose men always married women who were exactly 8% shorter. What would the correlation between their heights be?

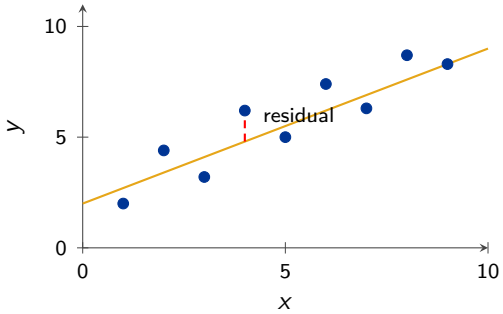
$$r = ?$$

# Simple Linear Regression: A Fitted Average Trend

## One quantitative predictor

$$\widehat{y} = b_0 + b_1 x$$

- $b_0$: fitted intercept ($x = 0$).
- $b_1$: predicted change in $y$ for 1 unit higher $x$.
- $\widehat{y}$: fitted *average* response, not every observed response.



# Ordinary Least Squares (OLS)

## How the line is chosen

For each observed pair, the residual is

$$e_i = y_i - \widehat{y}_i$$

OLS chooses $b_0$ and $b_1$ to minimize the sum of squared errors:

$$\mathrm{SSE} = \sum_{i=1}^{n} e_i^2.$$

A line can be fitted to curved data; that does **not** mean the line describes the pattern well.

# Residual Size and Model Fit

## Root mean squared error

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

## Coefficient of determination

$$R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}$$

$$\mathrm{SST} = \sum_i (y_i - \bar{y})^2$$

$R^2$ is improvement in squared fitting error versus predicting $\bar{y}$ for everyone.

# $R^2$ Is Not the Same as Correlation

**For simple OLS with an intercept**

$$R^2 = r^2$$

For example,

$$r = -0.80 \quad \Rightarrow \quad R^2 = 0.64$$

**Correlation ($r$)**

- Measures the strength and direction of a linear association.
- Ranges from $-1$ to 1.

**Coefficient of determination ($R^2$)**

- Measures the fraction of sample $y$ variation explained by the fitted line.
- Ranges from 0 to 1 for OLS with an intercept.

# Regression to the Mean

## Standard-unit form

$$\widehat{z}_y = r\, z_x$$

$$\widehat{y} = \bar{y} + r\frac{s_y}{s_x}(x - \bar{x})$$

If $|r| < 1$, the fitted standardized value is closer to zero than the predictor's standardized value.

## Example

For two comparable exams with $r = 0.5$, students scoring 2 SD above average on Exam 1 have fitted mean Exam 2 score 1 SD above average.

Improvement after selection on an extreme measurement does *not* by itself establish an intervention effect.

# The Sophomore Slump Example

## Baseball performance across seasons

Every year, baseball's major leagues honor their outstanding first-year players with the title "Rookie of the Year." The overall batting average for the Rookies of the Year is around .290, far above the major league batting average of .260. However, Rookies of the Year don't do so well in their second year—their overall second-season batting average is only .275. Baseball writers call this "sophomore slump," the idea being that star players get distracted by outside activities like product endorsements and television appearances. Do the data support the idea of the sophomore slump? Answer yes or no, and explain briefly.

# Why Association Is Not Causation

## Observational study

Investigators measure what naturally occurs; treatments/exposures are *not* assigned by the study.

## Potential explanations

- $X \longrightarrow Y$
- $Y \longrightarrow X$
- $Z \longrightarrow X$ and $Z \longrightarrow Y$ (confounding)
- Coincidence or selection effects

## Controlled experiment

Compare treatment and control under a defined protocol. Random assignment helps balance both measured and unmeasured confounders *on average*.

Placebos and blinding may reduce expectation and measurement biases when feasible

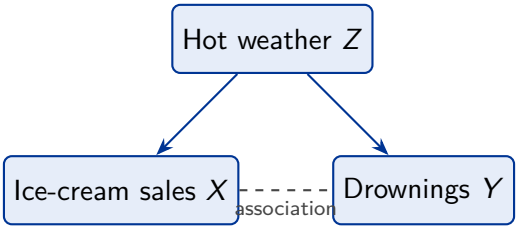
# Freedman Check 8: Did Randomization Work?

## Baseline smoking in two trials

Two clinical trials on prevention of heart attacks report baseline data on smoking, shown below. In one of these trials, the randomization did not work. Which one, and why?

| Trial | Group | Number of persons | Percent who smoked |
|---|---|---|---|
| (i) | Treatment | 1,012 | 49.3% |
| (i) | Control | 997 | 69.0% |
| (ii) | Treatment | 995 | 59.3% |
| (ii) | Control | 1,017 | 59.0% |

# Confounding: Draw the Story




**Interpretation**

An observed relation between $X$ and $Y$ may reflect common causes. A correlation coefficient alone cannot distinguish these stories.

# Boot Camp and Selection

## A voluntary program

California is evaluating a new program to rehabilitate prisoners before their release; the object is to reduce the recidivism rate—the percentage who will be back in prison within two years of release. The program involves several months of "boot camp"—military-style basic training with very strict discipline. Admission to the program is voluntary. According to a prison spokesman, "Those who complete boot camp are less likely to return to prison than other inmates."

(a) Is the prison spokesman's comparison based on an observational study or a randomized controlled experiment?

(b) True or false: the data show that boot camp worked.

# Simpson's Paradox and Group Composition

**Simpson's Paradox**

A trend in the overall data can **reverse** when the data are separated into groups.

- Groups may have different baseline rates.
- Groups may contribute different proportions to the overall data.
- Combining groups can hide or distort within-group trends.

Examine relevant groups before interpreting an overall trend.

# Sampling: Parameters and Random Statistics

## Notation

| Quantity | Population | Sample |
|---|---|---|
| Mean | $\mu$ | $\bar{x}$ |
| SD | $\sigma$ | $s$ |
| Variance | $\sigma^2$ | $s^2$ |

- $X$: random observation; $x$: realized value.
- $\bar{X}$: random sample mean across possible samples.
- $\bar{x}$: numerical mean computed from one observed sample.

## Think of repetitions

Draw a sample of size $n$, compute its average, and repeat. The resulting histogram of sample means is a **sampling distribution**.

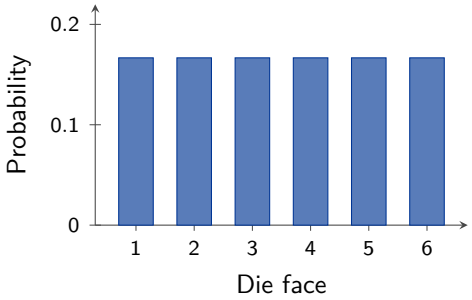
# The Fair Die: A Population We Know

## One independent fair roll

$$\mu = E(X) = \frac{1+2+3+4+5+6}{6} = 3.5$$

$$\sigma^2 = \frac{35}{12}, \qquad \sigma \approx 1.708$$

- A roll is uniform, not normal.
- These are *population* parameters for a known six-outcome distribution.



# Law of Large Numbers versus CLT

## Law of Large Numbers (LLN)

With sufficiently many independent observations under appropriate conditions, the sample average tends toward $\mu$.

## Central Limit Theorem (CLT)

For many independent observations from the same distribution with finite variance, the *distribution of sample averages* becomes approximately normal as $n$ grows.

## Key distinction

Individual data can be skewed or discrete while averages of sufficiently large samples are approximately normal.

# Expected Value and Standard Error of a Mean

**If $X_1, \ldots, X_n$ are independent with common mean $\mu$ and SD $\sigma$**

$$E(\bar{X}) = \mu, \qquad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

**Interpretation**

- **SD**: spread of individual outcomes.
- **SE**: spread of *sample averages* across repeated samples.
- Larger $n$ reduces SE, not the population SD.

**36 die rolls per sample**

$$E(\bar{X}) = 3.5$$

$$\mathrm{SE}(\bar{X}) = \frac{1.708}{\sqrt{36}} \approx 0.285$$

# Freedman Check 10: More Rolls, More Precision?

## Predict the total number of spots

A die will be thrown some number of times, and the object is to guess the total number of spots.

There is a one-dollar penalty for each spot that the guess is off. For instance, if you guess 200 and the total is 215, you lose $15. Which do you prefer: 50 throws, or 100? Explain.

# In Real Data, the Population SD Is Unknown

## Known population (rare)

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

This is exact under the independent common-variance sampling model.

## Estimated from a sample (usual)

$$\widehat{\mathrm{SE}}(\bar{X}) = \frac{s}{\sqrt{n}}$$

Replace unknown population SD $\sigma$ with sample SD $s$.

## What does $\pm$ mean?

"Mean $\pm$ something" could mean sample SD, SE, or a CI margin of error.

# From an Estimate to a Confidence Interval

## Estimate $\mu$ from $\bar{x}$

$$\text{CI} = \bar{x} \pm \text{margin of error}$$

- Point estimate: $\bar{x}$.
- Uncertainty of estimate: SE.
- Margin of error: critical value $\times$ estimated SE.

## Normal shortcut

For a sufficiently large independent sample, a roughly 95% CI is

$$\bar{x} \pm 2\widehat{\text{SE}}.$$

More generally, use an appropriate $z^*$ or $t^*$.

CI estimates the **population mean**, not the range of individual measurements.

# Why Student's $t$?

**When $\sigma$ is known**

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Under a normal mean/sampling model, use a normal critical value $z^*$.

**When $\sigma$ is estimated**

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

For independent, normal observations: $T \sim t_{n-1}$.

- Estimated denominator introduces extra uncertainty.
- $t$ has heavier tails than the normal.
- Degrees of freedom: $df = n - 1$.
- As $n$ grows, $t$ approaches normal.

# Confidence Interval Formula and Steps

## Population mean $\mu$, population SD unknown

$$\bar{x} \pm t_{1-\alpha/2,n-1} \frac{s}{\sqrt{n}}$$

1. Calculate $\bar{x}$, $s$, and $n$.
2. Estimated SE $= s/\sqrt{n}$.
3. Find $t^*$ for the target level and $df = n - 1$.
4. Margin of error $= t^* s/\sqrt{n}$.
5. Report lower/upper limits in the response units.
6. Explain the population and assumptions.

# What Does a CI Cover?

## Household size survey

A survey organization takes a simple random sample of 625 households from a city of 80,000 households. On the average, there are 2.30 persons per sample household, and the SD is 1.75. Say whether each of the following statements is true or false, and explain.

(a) The SE for the sample average is 0.07.

(c) A 95%-confidence interval for the average household size in the city is 2.16 to 2.44.

(d) 95% of the households in the city contain between 2.16 and 2.44 persons.

# What 95% Confidence Actually Means

**Repeated-sampling interpretation**

If we repeatedly draw samples by the same valid sampling method and construct 95% intervals, approximately 95% of those intervals will contain the *fixed* population mean $\mu$.

- The population mean does not change across samples.
- The calculated endpoints *do* change.
- One particular interval either covers $\mu$ or it does not.

**Not correct**

- "95% of individual values lie in this interval."
- "There is a 95% frequentist probability that the fixed $\mu$ is in this calculated interval."

# Freedman Check 12: Sample or Entire Class?

## Confidence interval—for what?

One term at the University of Pittsburgh, 400 students took the final in Statistics. Their scores averaged 65.3 out of 100, and the SD was 25. Now

$$\frac{25}{\sqrt{400}} = 1.25.$$

Is $65.3 \pm 2.5$ a 95%-confidence interval? If so, for what? If not, why not?

# Simpson's Paradox

## A hypothetical university

A hypothetical university has two departments, A and B. There are 2,000 male applicants, of whom half apply to each department. There are 1,100 female applicants: 100 apply to department A and 1,000 to department B. Department A admits 60% of the men who apply and 60% of the women. Department B admits 30% of the men who apply and 30% of the women. "For each department, the percentage of men admitted equals the percentage of women admitted; this must be so for both departments together." True or false, and explain briefly.

# Interpreting a Slope

## Height and weight

For men age 18–24 in the HANES5 sample, the regression equation for predicting height from weight is

$$\text{predicted height} = (0.0267 \text{ inches per pound}) \times (\text{weight}) + 65.2 \text{ inches}.$$

(Height is measured in inches and weight in pounds.) If someone puts on 20 pounds, will he get taller by

$$20 \text{ pounds} \times 0.0267 \text{ inches per pound} \approx 0.5 \text{ inches?}$$

If not, what does the slope mean?

# ‘Twice as Linear”?

**Think – Pair – Share: 2 min**

## A true-or-false claim

In one class, the correlation between scores on the final and the midterm was 0.50, while the correlation between the scores on the final and the homework was 0.25.
True or false, and explain: the relationship between the final scores and the midterm scores is twice as linear as the relationship between the final scores and the homework scores.

# Optional Freedman: Check the Normal Assumption

**Think – Pair – Share: 2 min**

## A bounded test score

Among applicants to one law school, the average LSAT score was about 169, the SD was about 9, and the highest score was 178. Did the LSAT scores follow the normal curve?

# Collections and Boolean Decisions

## Which data structure?

- **List**: ordered and mutable.
- **Tuple**: ordered and immutable.
- **Dictionary**: keys map to values.
- **Set**: unique members.

## Branch on a condition

```
reading = 23.5
limit = 20.0
if reading > limit:
    status = "high"
else:
    status = "within limit"
```

Python indentation marks a block; comparisons produce Booleans.

# Loops, Functions, and Vectorization

## A loop (element by element)

```
values = [2, 4, 6]
squares = []
for value in values:
    squares.append(value**2)
```

## A reusable function

```
def c_to_f(temperature_c):
    return 1.8 * temperature_c + 32
```

## NumPy (vectorized)

```
import numpy as np
x = np.array([2, 4, 6])
squares = x**2
mask = x > 3
selected = x[mask]
```

# Data Cleaning: What Can Change the Answer?

- Missing values: which rows are excluded, and why?
- Measurement units: seconds versus minutes; % versus fractions.
- Data types: numeric measurements versus categories.
- Duplicates, impossible values, recording errors.
- Averages across groups: group sample sizes can differ.

A sensor dataset has `NaN` for some temperatures. Is replacing every missing value with zero a neutral operation?

# Python Check: Predict the Result

## Given

```
x = np.array([2, 4, 6, 8])
mask = x >= 6
```

1. What values are in `x[mask]`?
2. What does `x.mean()` return?
3. Why might a `groupby()` mean differ from the overall mean?

## Useful commands

| Task | pandas / NumPy |
|---|---|
| Rows/columns | `.loc[rows, cols]` |
| Sort rows | `.sort_values()` |
| New column | `.assign()` |
| Group means | `.groupby().mean()` |
| Describe data | `.describe()` |

# Choose a Graph That Answers the Question

| Question | Display | Look for |
|---|---|---|
| One quantitative variable | Histogram / density plot | Shape, tails, modes, skew |
| Distribution by group | Side-by-side boxplots | Median, IQR, unusual points |
| Two quantitative variables | Scatterplot | Direction, form, strength, outliers |
| Measurement over time | Time-series line plot | Trend, jumps, periodicity |
| Groups with unequal counts | Grouped plots / summary | Within-group vs pooled patterns |

**Remember Anscombe's quartet**

Nearly identical means, SDs, and correlations can hide **very different** datasets.

# Summaries Are Not Enough

**Dataset A**

2, 3, 4, 5, 6

**Dataset B**

2, 3, 4, 5, 26

1. Compare the medians.
2. Compare the means.
3. What happens to sample SD?
4. What would a boxplot help you notice?

# SciPy: Probability versus Percentile

## Model with mean 10, SD 2

```
from scipy import stats
model = stats.norm(loc=10, scale=2)
p_left = model.cdf(12)
p_right = model.sf(12)
cutoff = model.ppf(0.95)
p_between = model.cdf(12) - model.cdf(8)
```

## Interpretation

- `cdf(x)`: area left of $x$.
- `sf(x)`: area right of $x$.
- `ppf(p)`: value with proportion $p$ below it.
- `pdf(x)`: density, *not* probability at a point.

# Normal-Approximation Check

**Question**

A manufacturing process produces highly right-skewed cycle times. A scientist averages $n = 100$ independent cycle times per batch.

1. Must the individual times be normal?
2. Why might batch averages be closer to normal?
3. Does the CLT correct systematic bias?

**Check assumptions**

- Independent, comparable observations
- Finite variation
- Adequately large $n$ for the shape

# Quick Check: Three Calibrations

**Scenario**

A reference length is **36.00 in**. A tool reports 35.96, 36.01, and 36.03 in.

1. What estimate would you use for the typical reading?
2. Does the spread of the readings alone establish accuracy?
3. If the tool always reads 0.20 in too high, what type of error is this?

# Correlation Under Unit Changes and Restrictions

## Invariance

- Shifting $x$ or $y$: **no change** in $r$.
- Multiplying either by a positive factor: no change.
- Multiplying just one by a negative factor: sign flips.
- Swapping $x$ and $y$: correlation stays the same.

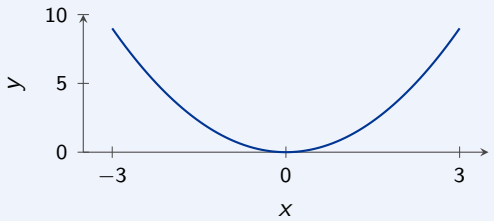
## But correlations can change when...

- We restrict the range of $x$ or $y$.
- We include/exclude an unusual observation.
- We pool or split subgroups.
- We replace individuals with group averages.

$r = 0$ **means no linear correlation**, not necessarily no relationship.

# Three Common Traps in Correlation

## A curve can have $r \approx 0$




## Do not overgeneralize

- **Ecological correlation**: relationship across group rates or averages; need not hold for individuals.
- **Outlier**: can create, erase, or reverse apparent linear correlation.
- **Confounding**: a third variable may drive both $x$ and $y$.

# Correlation Check: True or False?

1. $r = 0.90$ means increasing $x$ by 1 unit increases $y$ by 0.90 units.
2. A negative correlation means $x$ causes $y$ to decrease.
3. Converting temperatures from Celsius to Fahrenheit changes $r$.
4. If $r = -1$, all observed pairs lie exactly on a decreasing straight line.
5. A strong curved relationship can have a small Pearson $r$.

# Interpret a Line, Check Extrapolation

**Example: An engineering experiment**

A fitted model for runtime (hours) versus battery capacity (Wh) is

$$\widehat{\text{runtime}} = 1.2 + 0.045\,\text{capacity}.$$

The observed battery capacities range from **50 to 100 Wh**.

1. Interpret the slope in units.
2. Predict runtime for 80 Wh.
3. Why is predicting for 150 Wh more uncertain?
4. Does this fitted line prove battery capacity *causes* the difference?

# Regression in Python: Two Approaches

## Slope and intercept

```
import numpy as np
slope, intercept = np.polyfit(
    x, y, deg=1)
yhat = intercept + slope * x
residual = y - yhat
```

## statsmodels model object

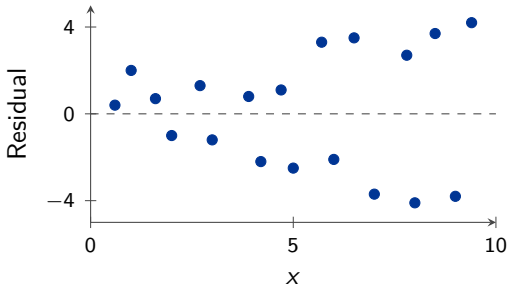
```
import statsmodels.formula.api as smf
fit = smf.ols("y ~ x", data=df).fit()
print(fit.params)
print(fit.rsquared)
residual = fit.resid
```

Both fit a straight-line mean trend with an intercept by default. **Inspect data and residuals before trusting the model.**

# Residual Diagnostics: Center, Spread, Shape

## What to examine

- **Center**: systematic curves or remaining trends?
- **Spread**: roughly constant width or a fan?
- **Shape**: roughly symmetric bell shape when Gaussian errors are assumed?
- **Unusual cases**: large residual, extreme $x$, or influential point?




Illustrative example: widening spread (heteroscedasticity).

# Unusual Regression Points: Three Ideas

| Concept | What is unusual? | What to check |
|---|---|---|
| Large residual | $y$ far above/below fitted line | Check measurement; inspect context |
| High leverage | $x$ far from most other $x$ values | Check predictor range; investigate |
| Influential point | Removing it substantially changes fit | Refit with/without; compare coefficients |

**Distinction**

High leverage is not automatically high influence. A large residual does not automatically change the slope substantially.

# Regression Check: Match the Diagnosis

1. Residuals follow a clear U-shaped pattern.
2. Residuals have narrow spread at low $x$, wide spread at high $x$.
3. A point has a very extreme $x$ but sits close to the fitted line.
4. Omitting one point changes the fitted slope dramatically.
5. An equation fitted as $y$ on $x$ is rearranged to predict $x$ from $y$.

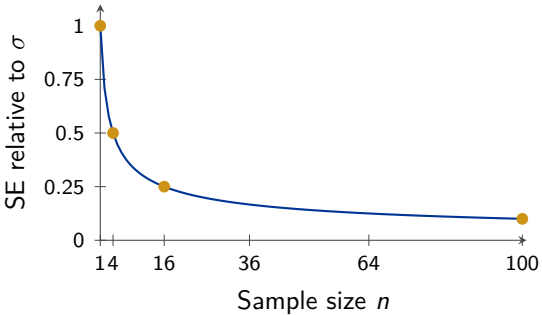
# Causation Check: Improve the Design

**Scenario**

In historical records, students attending optional tutoring have lower mean grades than students who do not attend.

1. Can we conclude tutoring reduces grades?
2. Name an important possible confounder.
3. What experimental or comparison design would improve the causal claim?

# Sample Size: How Fast Does Precision Improve?




**Scaling law**

$$\mathrm{SE} \propto \frac{1}{\sqrt{n}}$$

- $4\times$ as many samples $\Rightarrow$ SE halves.
- $9\times$ as many $\Rightarrow$ SE divided by 3.
- No sample size automatically fixes selection bias or confounding.

# SE Practice: Individual Variation versus Precision

**Scenario**

A random sample of $n = 64$ component lifetimes has sample mean $\bar{x} = 120$ hours and sample SD $s = 24$ hours.

1. What estimates the spread of individual lifetimes?
2. Estimate the standard error of the sample mean.
3. If $n$ were 256, with $s$ still 24 hours, what would the estimated SE be?

# Two-Sided Confidence Levels and Critical Values

## Tail areas

$$\text{confidence level} = 1 - \alpha$$

$$t^* = t_{1-\alpha/2,\, n-1}$$

Here the subscript denotes the *left-tail quantile*. Equivalently, some textbooks use $t_{\alpha/2,n-1}$ to mean an *upper-tail* cutoff; define the convention.

## 95% critical values

| $n$ | $df$ | $t^*$ |
|---|---|---|
| 5 | 4 | 2.776 |
| 10 | 9 | 2.262 |
| 20 | 19 | 2.093 |
| 30 | 29 | 2.045 |
| 100 | 99 | 1.984 |
| Large | — | $\approx 1.96$ |

# Small Sample Confidence Interval

**Printed-part tensile strength**

A random sample of $n = 10$ parts has mean strength $\bar{x} = 22.3$ MPa and sample SD $s = 3.8$ MPa. Construct a 95% CI for the population mean.

1. Estimate SE and determine degrees of freedom.
2. Use $t^* = 2.262$ for $df = 9$.
3. Compute the margin of error and endpoints.

# When Can We Trust the Mean CI?

## Sampling and distribution

- Representative random sample or credible sampling design.
- Independence (or a justified dependence adjustment).
- For small $n$, approximately normal underlying data and no major outliers.
- For large $n$, CLT may justify a mean approximation even with nonnormal individual data.

## What the formula cannot repair

- Selection bias
- Systematic measurement bias
- Confounding or causal ambiguity
- Strong dependence ignored in the SE calculation

# SciPy Confidence Interval Reference

## When population SD is unknown

```
from scipy import stats
n = len(data)
xbar = data.mean()
se = stats.sem(data)
alpha = 0.05
t_star = stats.t.ppf(1-alpha/2, df=n-1)
ci = (xbar-t_star*se, xbar+t_star*se)
```

## Interpret the functions

- `sem()`: sample-based SE of a mean.
- `t.ppf()`: critical value from cumulative area.
- `t.interval()`: direct interval alternative.

# Synthesis: Choose the Right Summary

An engineer analyzes 80 component lifetimes and a temperature predictor.

| Question | Useful approach |
|---|---|
| How variable are individual lifetimes? | Sample SD, IQR, histogram |
| How precise is the estimated mean lifetime? | Estimated SE, $s/\sqrt{n}$ |
| What range estimates the population mean? | Mean CI with $t^*$ and assumptions |
| Is lifetime associated with temperature? | Scatterplot; Pearson $r$; fitted line |
| How well does a line capture the trend? | Residuals; RMSE; $R^2$ |
| Does temperature *cause* a change? | Evaluate experimental design/confounders |

# More Practice: Explain the Reasoning

1. A histogram is right-skewed with an extreme high value. Which center/spread summaries are more resistant?
2. A normal model has $\mu = 100$ and $\sigma = 10$. What is the $z$-score of 85?
3. A regression has slope $-2.4$ and $R^2 = 0.49$. What is the sign of its simple-regression correlation?
4. An estimated SE is 4. What happens to it if sample size is multiplied by 4, with similar SD?
5. If the population SD is unknown and $n = 12$, what distribution and df are used for a conventional mean CI under normal sampling assumptions?

# Formula Reference

## Describe and standardize

$$\bar{x} = \frac{1}{n}\sum_i x_i, \quad s^2 = \frac{\sum_i (x_i - \bar{x})^2}{n-1}$$

$$\mathrm{IQR} = Q_3 - Q_1, \quad z = \frac{x - \mu}{\sigma}$$

## Regress

$$\hat{y} = b_0 + b_1 x, \quad e_i = y_i - \hat{y}_i$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i e_i^2}{n}}, \quad R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}$$

## Sample mean and CI

$$E(\bar{X}) = \mu, \quad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

$$\widehat{\mathrm{SE}}(\bar{X}) = \frac{s}{\sqrt{n}}$$

$$\text{mean CI} = \bar{x} \pm t_{1-\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

## Normal areas

$$P(a < X < b) = F(b) - F(a)$$